Introduction to Software Engineering

Software Requirements Specification

The student team is required to complete the **Software Requirements Specification** documentation for the assigned course project, following the attached template.

Software Engineering Department
Faculty of Information and Technology
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Software Requirements Specification

Objectives

This document focus on the following topics:

✔️ Completing the Software Requirements Specification (SRS) document with the following sections:

- Problem Statement
- Stakeholders
- Functional, non-functional and domain requirements
- Use case model and use case specifications
- Prototype / mockups
- Traceability and requirements review

✔️ Including an AI usage log for this phase

✔️ Understanding the Software Requirements Specification (SRS) document.

1 Member Contribution Assessment

[List every member. Contribution (%) must add up to 100%. Evidence: links to tasks on the board, commits or pull requests.]

ID	Name	Role(s)	Contribution (%)	Evidence
24127352	Nguyễn Lương Hoàng Duy	Team lead	25%	Task board
24127436	Niê Ngọc Lâm	BA, Developer	25%	Task board
24127521	Nguyễn Việt Quang	Tester	25%	Task board
24127343	Trần Phúc Đạt	Designer	25%	Task board

2 Introduction

2.1 Purpose and Scope

Purpose:

- **Establishing Alignment and a Single Source of Truth:** This document creates a clear, shared understanding among stakeholders, product managers, developers, UI/UX designers, and QA teams. By documenting expectations explicitly, it prevents misunderstandings about feature behavior early.
- **Defining System Scope and Preventing Out Of Scope:** By outlining explicit functional requirements (features, inputs, outputs) and non-functional requirements (security, scalability, performance), an SRS establishes fixed project boundaries. It clarifies both what is included and what is explicitly out of scope.
- **Guiding Architecture and System Design:** Software architects and developers rely on the SRS to make informed decisions regarding system architecture, tech stack selection, database schema design, and API integrations based on concrete parameters like response times, user load, and security protocols.
- **Providing a Baseline for Testing and Validation:** The QA team uses the SRS to write test plans, test cases, and user acceptance criteria. A requirement is valid only if it is testable, the SRS provides the benchmark against which the final product is verified.
- **Enabling Accurate Resource, Cost, and Timeline Estimation:** Project managers use detailed specifications to break down tasks, calculate required development hours, allocate engineering resources, and create realistic project roadmaps and budgets.
- **Serving as a Legal and Contractual Agreement:** In vendor-client relationships or outsourcing scenarios, the SRS often forms part of the legally binding contract. It acts as the agreed-upon criteria for project sign-off and milestone payments.

Product Scope:

The followings are the in scope main features in accordance with the MoSCoW priority:

ID	Feature	Description	Priority
F01	Task Creation and Editing	Allows users to create, edit, delete, and mark daily to-do tasks as completed.	Must
F02	Cross-Platform Local Storage (Offline-First)	Automatically saves tasks and user settings locally on the device (using SQLite), ensuring the app works seamlessly even without an internet connection.	Must
F03	Priority and Color Coding	Enables users to categorize tasks by priority level (e.g., High, Medium, Low) with visual color tags.	Must
F04	Quick Search & Keyword Filtering	Allows users to instantly search through their entire task list using keywords.	Could
F05	Daily Task List & Status Filtering	Displays daily tasks cleanly in a unified UI and allows filtering by status , priority or category.	Must
F06	Cloud Database Synchronization	Automatically syncs task data between the user's PC and mobile device via a cloud database backend (e.g., Supabase) when online.	Should
F07	Visual Task Progress Tracking	Provides a dynamic progress bar or completion percentage reflecting	Must

		daily task progress to keep users motivated.	

Out of scope features:

- Real-time multi-user collaboration: Sharing tasks and collaboration between many individuals.

Intended readers:

- Technical and Engineering teams
- QA and Testing teams
- Business Management and Stakeholders
- Operations and Legal teams

2.2 Glossary

Term	Definition
Functional requirements	State how the system should react to particular inputs and how the system should behave in particular situations.
Non-Functional requirements	State constraints on services or functions offered by the system,
Domain requirements	Functional or Non-Functional requirements that are derived from the specific domain that the application is used in. Reflects the characteristics of that domain.
Stakeholders	Any person or organization who is affected by the system.
Use case	A use case is a set of actions or event scenarios that defines how an actor interacts with a system to achieve a single, completed outcome.
Actors	An actor represents a role played by a user, an external system, a hardware device, or a time-based event that interacts directly with the software system.
MoSCoW prioritization	A technique used in software development, product management, and business analysis to categorize and rank requirements based on their necessity to a project's success or release scope.

3 Problem Statement

[Elaborate the preliminary problem statement from PA1: business context, current problems, goals, operating environment, design and implementation constraints.]

Business Context

In modern academic and professional settings, individuals frequently manage overlapping responsibilities within tightly scheduled days. Students must balance lectures, assignments, exam preparation, group projects, and personal routines. Concurrently, office workers and freelancers need to organize daily deliverables, meet project deadlines, and sustain personal productivity. To maintain focus and prevent burnout, these user groups require a clear, reliable system to outline, track, and evaluate their daily workload without friction.

Current Problems and Limitations

Despite the abundance of task organization tools, current solutions present many drawbacks for daily planning:

- **Mental Notes:** Unreliable and prone to human error, leading to forgotten tasks, missed deadlines, and cognitive overload.
- **Handwritten Lists:** Depending on one's ability to present and organize the list. A handwritten list can be messy and not visually appealing.
- **Enterprise Productivity Platforms:** Feature-heavy applications (e.g., Jira, Notion, Asana) introduce high configuration friction, complex interfaces, and unnecessary learning curves for users who only need quick daily task tracking.
- **Mandatory Cloud Dependency:** Many existing tools require continuous internet connectivity and cloud account creation, creating barriers for continuous offline use and raising data privacy concerns for users who prefer local data ownership.

Goals and Proposed Solution

The proposed product, **WhatYaDoing?**, resolves these challenges by delivering a lightweight, intuitive, cross-platform task management application designed for daily productivity and clear progress visibility.

Key objectives of the solution include:

- **Core Task Lifecycle Management:** Streamlining daily task creation, modification, completion, and deletion through a straightforward interface.
- **Visual Prioritization and Filtering:** Implementing color-coded priority levels and flexible filtering views (by completion status and priority tier) to keep high-urgency items front and center.
- **Progress Overview:** Providing visual progress indicators (such as completion percentages or status bars) to give immediate daily feedback.
- **Hybrid Data Architecture:** Supporting offline-first local storage for immediate access while syncing data across devices when network connectivity is available.

Operating Environment

- **Deployment Platforms:** Cross-platform deployment initially targeting **Windows (Desktop)** and **Android (Mobile)**.
- **User Interface Layer: React Native** for cross-platform UI components, styled with **NativeWind** to ensure responsive layouts across desktop and mobile screens.
- **Backend & API Layer: Node.js/Express** written in **JavaScript and TypeScript**, serving core application logic and REST API endpoints.
- **Data Storage & Synchronization: Supabase** as the primary backend-as-a-service for cloud database management and cross-device sync, paired with **Local Storage** for offline persistence.
- **Build Pipeline & Testing: GitHub Actions** for CI/CD automation and application packaging; **Jest** for automated unit and integration testing.

Resource and Schedule Constraints:

- Developed by a team of four students within a 3-week core implementation window, requiring strict prioritization of essential features over complex architectures.
- **Data Synchronization & Conflict Handling:** The application must handle offline state persistence, reconnection state handling, conflict resolution, and invalid data gracefully to prevent data loss or corruption.
- **Scope Boundaries:** Multi-user collaboration, granular permission management, and third-party calendar integrations (e.g., Google Calendar, Outlook) are explicitly excluded from the initial release scope.
- **Course & Engineering Policy:** Development must comply with strict software engineering practices, including formal code reviews, maintainable project documentation, automated testing coverage, and transparent documentation of AI tool usage.

4 Stakeholders

[List the stakeholders (or draw a context diagram) and explain their roles and interests.]

#	Stakeholder	Description	Interests / concerns
1	Course Instructor & TA	Course instructors and teaching assistants evaluating the project	Evaluates how well the team follows the principles of the software development waterfall model throughout the course of the project.
2	Project Sponsor	The TA will assume the role of a project sponsor in this project	Give approval or suggestions on documents such as the project proposal. As the project sponsor, give the final judgement on whether the end product delivers what was promised in the proposal.
2	Project Team Members (The 4 idiots)	Developers, designers, tester, and team lead building the application	Completing the project on time, meeting work standards, fair contribution, achieving target grade (7-8), and maintaining code quality.

3	Students (End User)	College or school students managing academic deadlines, coursework, and daily personal chores	Free, fast, local task tracking, easy assignment management, and visual progress tracking to balance academic workload.
4	Office Workers & Freelancers (End User)	Professionals needing a clean, offline tool for immediate daily task dumping and priority management	High privacy (local storage), zero account setup, intuitive interface, priority color-coding, and offline availability.

5 Requirements

5.1 Functional Requirements

[Use "The system shall …" statements. Number the requirements and group them by feature. Give each a MoSCoW priority and a source (stakeholder, interview, document).]

ID	Requirement	Feature	Priority	Source
FR-01	The system shall allow the user to create a task with a required title, a required category, and optional description, priority and date; if no date is given, the task date shall default to today	F01	Must	PA1, 2.4
FR-02	The system shall reject creating a task with an empty title or without a category and show an error message	F01	Must	PA1, 2.4
FR-03	The system shall allow the user to edit the title, description, priority, category and date of an existing task	F01	Must	PA1, 2.4
FR-04	The system shall allow the user to delete a task after the user confirms the deletion	F01	Must	PA1, 2.4
FR-05	The system shall allow the user to mark a task as completed and to revert it to active	F01	Must	PA1, 2.4
FR-06	The system shall save every task change (create, edit, delete, complete) to local storage automatically, without a manual save action	F02	Must	PA1, 2.4
FR-07	The system shall load all saved tasks from local storage when the application starts	F02	Must	PA1, 2.4
FR-08	The system shall allow all Must-have features to be used without an Internet connection	F02	Must	PA1, 2.4
FR-09	The system shall allow the user to	F03	Must	PA1, 2.4

	assign priority level (High, Medium, Low) and Category (Work, health, personal) to each task, with Medium and Work as the default			
FR-10	The system shall display each task with a color tag corresponding to its priority level	F03	Must	PA1, 2.4
FR-11	The system shall display the list of tasks of the selected day	F05	Must	PA1, 2.4
FR-12	The system shall allow the user to switch between days to view their tasks	F05	Must	PA1, 2.4
FR-13	The system shall allow the user to filter tasks by status (All, Active, Completed)	F05	Must	PA1, 2.4
FR-14	The system shall allow the user to filter tasks by priority (High, Medium, Low)	F05	Must	PA1, 2.4
FR-15	The system shall allow the user to filter tasks by category	F05	Must	Team decision
FR-16	The system shall apply the Status, Priority and Category filter together when they are selected	F05	Must	Team decision
FR-17	The system shall display the daily progress as a percentage (completed tasks / total tasks of the day) and as a progress bar	F07	Must	PA1, 2.4
FR-18	The system shall update the progress bar and percentage immediately when a task is completed, reverted, added or deleted	F07	Must	PA1, 2.4
FR-19	The system shall allow the user to search tasks by a keyword in the title	F04	Could	PA1, 2.4
FR-20	The system shall display only tasks matching the keyword searched	F04	Could	PA1, 2.4
FR-21	The system shall allow the user to sign in to an account in order to use cloud synchronization	F06	Should	Team decision
FR-22	The system shall synchronize local task changes to the cloud database when an Internet connection is available	F06	Should	PA1, 2.4
FR-23	The system shall keep changes made offline in a queue and synchronize them	F06	Should	Team

	automatically when the connection is restored			decision
FR-24	The system shall resolve synchronization conflicts using the most recently updated version of a task (last-write-wins)	F06	Should	Team decision
FR-25	The system shall allow the user to change the date format in the settings. The default format shall be DD-MM-YYYY.	F01	Should	Team decision

5.2 Non-Functional Requirements

[Every non-functional requirement must be verifiable: state a metric and the conditions. Example: "Under 200 concurrent users, the search page shall return results in ≤ 1 s at P95."]

ID	Category	Requirement (measurable)	How it will be verified
NFR-01	Performance	With 500 stored tasks on a PC with 8 GB RAM, the application shall start and display the task list in ≤ 3 s	Measure startup time 5 times with a seeded dataset; the average must be ≤ 3 s
NFR-02	Security	If the user signs in, the user shall be able to read and modify only their own tasks (100% of unauthorized access attempts rejected); no API key or secret shall appear in client code or the repository	Test with 2 accounts (access to the other's data returns an error); scan the repository for secrets before release
NFR-03	Usability	A first-time user shall be able to create their first task and mark it completed in ≤ 2 minutes without instructions	Usability test with ≥ 5 students; ≥ 4 of 5 succeed within the time limit
NFR-04	Usability	Given a list of tasks, a first-time user shall be able to filter the task according to certain keywords or status in less than 2 minutes without instructions	Usability test with ≥ 5 students; ≥ 4 of 5 succeed within the time limit
NFR-05	Performance	With up to 1,000 tasks, applying a filter or keyword search shall update	Performance test with a 1,000-task dataset,

		the list in ≤ 300 ms	timed over 10 runs
NFR-06	Performance	After the user performs create/edit/delete/complete, the UI shall reflect the change in ≤ 200 ms	Automated UI test that measures the time from the action to the updated screen, over 10 runs
NFR-07	Reliability	No task data shall be lost when the application is force-closed immediately after a change (changes are persisted within 1 s)	Perform 20 create/edit actions, force-close the app, reopen it and verify all data is present
NFR-08	Reliability	With the network disabled, all Must-have features shall run with 0 failures	Run all Must-have test cases in airplane mode on Windows and Android
NFR-09	Reliability	After the connection is restored, offline changes shall be synchronized with no duplicated or lost tasks (≥ 20 changes tested);	Make 20 changes offline, reconnect, then compare the data on both devices
NFR-10	External (Privacy)	The system shall store task data only on the user's device and in the application's database,	Inspect network traffic during a test session; review the data flow in the C4 diagram
NFR-11	Portability	The application shall run on Windows 10/11 and Android 10 (API 29) or later	Install the packaged release on at least 1 Windows machine and 1 Android device/emulator, then run the smoke test

5.3 Domain Requirements and Business Rules

ID	Rule	Source (policy, law, standard)
BR-01	A task title is required and shall contain at most 100 characters; the description is optional and shall contain at most 500 characters	Team decision
BR-02	A task has exactly one status at a time: Active or Completed	Team decision

BR-03	The priority levels are fixed to High, Medium and Low, shown in red (High), yellow (Medium) and green (Low)	Team decision
BR-04	A task has exactly one category at a time	Team decision
BR-05	A task belongs to exactly one day, determined by its date	Team decision
BR-06	The user can choose the format in which dates are displayed (DD/MM/YYYY, MM/DD/YYYY or YYYY-MM-DD); the default format is DD/MM/YYYY, and changing the format does not change the stored task dates	Team decision
BR-07	Daily progress = (number of Completed tasks of the day ÷ total number of tasks of the day) × 100, rounded to the nearest whole percent; deleted tasks are not counted	Team decision
BR-08	Keyword search covers the tasks of all days and is case-insensitive	Team decision
BR-09	Each task has a unique ID and a last-updated timestamp; when two versions of the same task conflict, the version with the later timestamp is the most recently updated one (applies only if F06 is implemented)	Team decision
BR-010	When the user deletes a task and cloud synchronization is enabled, the task shall also be removed from the cloud database (applies only if F06 is implemented)	Team decision

5.4 Acceptance Criteria

[For each Must-have feature, write acceptance criteria in Given / When / Then form. They will become acceptance test cases in PA4.]

F01 – Task Creation and Editing

AC-01 (FR-01)

Given the user is on the Create Task screen,

When the user enters a valid title, selects a date and saves,

Then the task appears in the list of that day with status Active.

AC-02 (FR-01, BR-05)

Given the user is on the Create Task screen and no date is selected,

When the user enters a valid title and saves,

Then the task date is today and the task appears in today's list.

AC-03 (FR-02)

Given the user is on the Create Task screen,

When the user leaves the title empty or does not choose a category and saves,

Then no task is created and an error message is shown.

AC-04 (FR-03)

Given a task exists in the list,

When the user changes its title and priority and saves,

Then the list shows the updated title and priority, and no duplicate task is created.

AC-05 (FR-04)

Given a task exists in the list,

When the user chooses to delete the task and confirms,

Then the task is removed from the list.

AC-06 (FR-04)

Given a task exists in the list,

When the user chooses to delete the task and cancels,

Then the task remains unchanged in the list.

AC-07 (FR-05)

Given an Active task exists,

When the user marks the task as completed,

Then the task status becomes Completed.

AC-08 (FR-05)

Given a Completed task exists,

When the user reverts the task,

Then the task status becomes Active.

F02 – Cross-Platform Local Storage (Offline-First)

AC-09 (FR-06, FR-07)

Given the user has created, edited and deleted a task,

When the user closes the application (including force-closing it right after a change) and opens it again,

Then all changes are still present and no manual save was needed.

AC-10 (FR-08, NFR-07)

Given the device has no Internet connection,

When the user creates, edits, deletes and completes a task, sets a priority, filters the list and views the progress,

Then all actions work without errors.

AC-11 (FR-24, BR-06)

Given the date display format is DD/MM/YYYY (default),

When the user selects YYYY-MM-DD in the settings and restarts the application,

Then all dates are displayed as YYYY-MM-DD and the dates of the tasks are unchanged.

F03 – Priority and Color Coding

AC-12 (FR-09)

Given the user is on the Create Task screen,

When the user saves a task without choosing a priority,

Then the task has priority Medium.

AC-13 (FR-10, BR-03, BR-04)

Given three tasks exist with priorities High, Medium and Low,

When the user views the task list,

Then the tasks show a red, a yellow and a green tag, each with the text label High, Medium or Low.

F05 – Daily Task List and Status Filtering

AC-14 (FR-11, FR-12, BR-05)

Given Day A has 2 tasks and Day B has 1 task,

When the user selects Day B,

Then only the 1 task of Day B is displayed.

AC-15 (FR-13)

Given the selected day has 2 Active tasks and 1 Completed task,

When the user selects the filter Completed, then Active, then All,

Then the list shows 1, 2 and 3 tasks respectively.

AC-16 (FR-14)

Given the selected day has one High, one Medium and one Low task,

When the user selects the priority filter High,

Then only the High task is displayed.

AC-17 (FR-15, BR-08)

Given the selected day has a High Active task, a High Completed task and a Low Active task,

When the user selects the status filter Active and the priority filter High,

Then only the High Active task is displayed.

AC-18 (FR-15, BR-04)

Given the selected day has one task in category A and one task in category B,

When the user selects the category filter A,

Then only the task in category A is displayed.

F07 – Visual Task Progress Tracking

AC-19 (FR-16, BR-07)

Given the selected day has 3 tasks and 1 of them is Completed,

When the user views the day,

Then the progress shows 33% and the progress bar is filled accordingly.

AC-20 (FR-17, BR-07)

Given the progress of the day is 33% (1 of 3 tasks Completed),

When the user marks another task as completed,

Then the progress shows 67% immediately.

AC-21 (FR-17, BR-07)

Given the selected day has 2 tasks and 1 of them is Completed (50%),

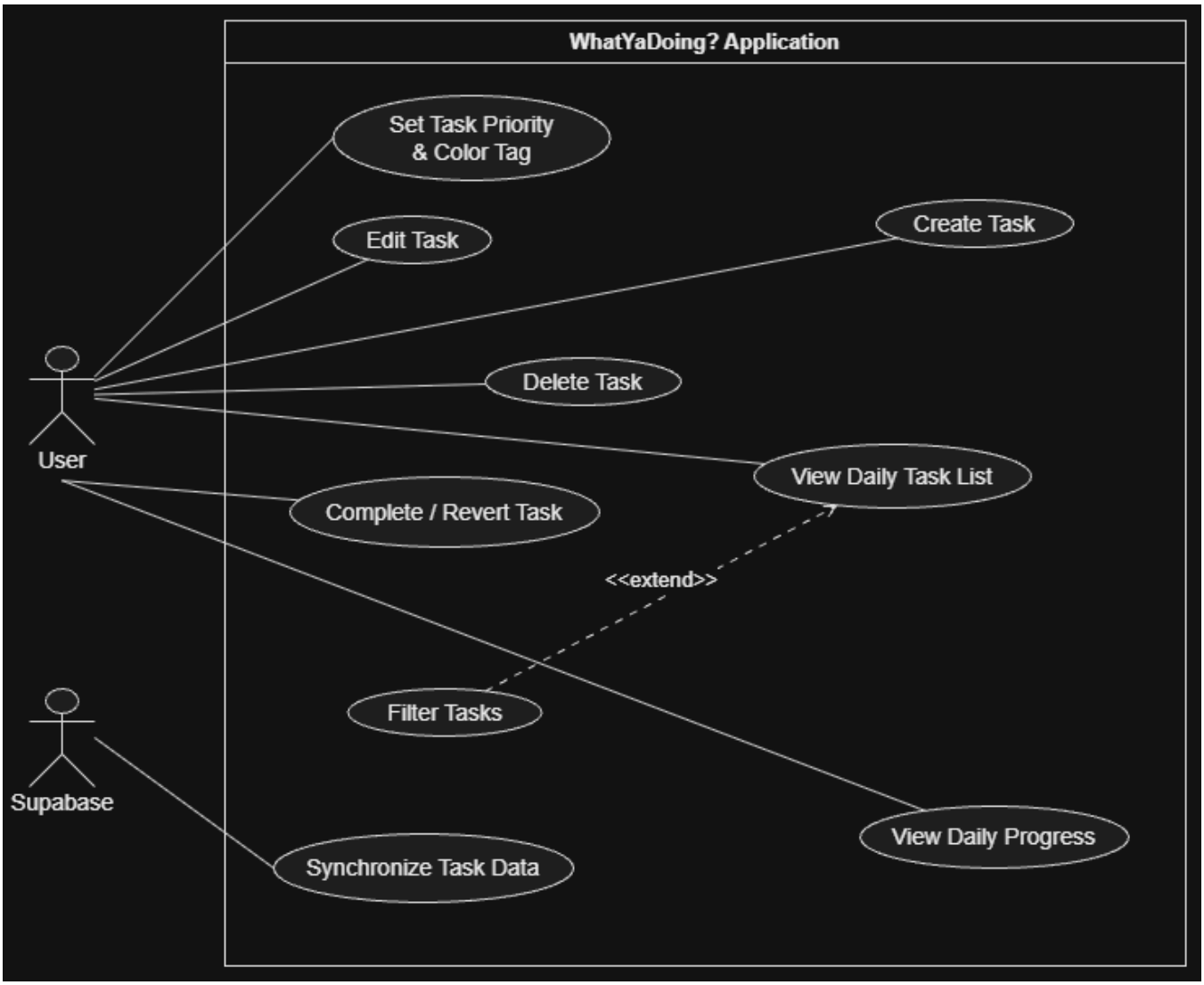
When the user deletes the Active task,

Then the progress shows 100% immediately.

6 Use Case Model

6.1 Use Case Diagram

[Draw the use case diagram of the system (StarUML, draw.io, PlantUML, Mermaid, ...). Use include / extend / generalisation correctly.]



6.2 Actors

Actor	Description
User	The primary user of WhatYaDoing? who creates, edits, deletes, completes, prioritizes, and filters tasks and views daily progress.
Supabase	An external service used to store and synchronize the user's task data when the application is online, allowing data to be accessed across supported devices.

6.3 Use Case List

ID	Use case	Primary actor	Priority	Related FR
UC-01	Create task	User	Must	FR-01
UC-02	Edit task	User	Must	FR-03
UC-03	Delete task	User	Must	FR-04
UC-04	Complete / Revert task	User	Must	FR-05
UC-05	Set Task priority	User	Must	FR-09
UC-06	View daily task list	User	Must	FR-11, FR-12
UC-07	Filter tasks	User	Must	FR-13, FR-14, FR-15, FR-16
UC-08	View daily progress	User	Must	FR-17, FR-18
UC-09	Synchronize Task data	User / Supabase	Should	FR-21 to FR-24

7 Use Case Specifications

[Specify the most important use cases (all Must-have use cases, at least 6). Number every step of the flow.]

7.1 Use Case UC-01

Use case ID	UC-01
Use case name	Create task
Brief description	Allows the user to create a new task with a title, description, and priority level, saving it locally and synchronizing it with the database when online.
Primary actor	User
Secondary actors	Supabase (Database Service) / Local Storage
Priority	Must
Related requirements	FR-01, FR-02, FR-06, FR-18, FR-21 to FR-24, NFR-06, BR01, BR-03 to BR-05
Trigger	The user clicks on the “Add Task”button in the main dashboard
Pre-conditions	1. The application is opened and displays the main dashboard. 2. (If authenticated) The user is signed into their active session.
Post-conditions	1. A new task record is created and stored in local storage and the Supabase database. 2. The main dashboard updates to display the new task and refreshes the daily progress bar.

Main flow	1. The user clicks the "Add Task" button on the main dashboard. 2. The system displays the "Create Task" dialog form. 3. The user enters the task title, optional notes (description), and selects a priority level. 4. The user clicks the "create task" button. 5. System validates the entered task data. 6. The system saves the task to local storage and syncs the record with Supabase if online. 7. The system closes the dialog form and updates the main dashboard task list.
Alternative flows	**AF-01: User Cancels Task Creation** 1. At step 3 or 4 of the main flow, the user clicks the "Cancel" or "Close" button. 2. The system discards all entered inputs. 3. The system closes the dialog form and returns to the main dashboard without updating the database/ local storage.
Exception flows	**EF-01: Required Fields Missing** 1. At step 5 of the main flow, the system detects that the task title is empty. 2. System highlights the task title field and displays an inline error message ("Task title is required"). 3. The system keeps the modal open for the user to correct the input. **EF-02: Offline Synchronization** 1. At step 6 of the main flow, the app is in offline mode. 2. The system saves the task directly to local storage, flags the item for pending sync, and updates the dashboard

	UI. 3. The system automatically syncs the local task to Supabase once online.
Business rules	1. BR-01: A task title is required and shall contain at most 100 characters; the description is optional and shall contain at most 500 characters 2. BR-03: The priority levels are fixed to High, Medium and Low, shown in red (High), yellow (Medium) and green (Low) 3. BR-04: A task has exactly one category at a time 4. BR-05: A task belongs to exactly one day, determined by its dat
Non-functional constraints	1. NFR-06: After the user performs create/edit/delete/complete, the UI shall reflect the change in ≤ 200 ms

7.2 Use Case UC-02

Use case ID	**UC-02**
Use case name	Edit task
Brief description	Allows the user to modify details of an existing task (title, description, priority, category, date) and save the changes locally/ database.
Primary actor	User
Secondary actors	Supabase/ local storage
Priority	Must
Related requirements	FR-03, FR-06, NFR-06, NFR-07
Trigger	The user selects an existing task and clicks the "Edit" icon on the task.

Pre-conditions	1. The application is open and displaying the main dashboard. 2. At least one task exists on the dashboard.
Post-conditions	1. Updated task details are saved to local storage and synced to Supabase. 2. The dashboard updates to display the edited task information.
Main flow	1. The user clicks the "Edit" icon of a specific task. 2. The system displays the "Edit Task" dialog form pre-filled with the current task details. 3. The user modifies the title, description, priority, category, or date. 4. The user clicks the "Save Changes" button. 5. The system validates the updated data against business rules. 6. The system saves updated details to local storage automatically. 7. The system closes the modal and updates the dashboard view.
Alternative flows	**AF-01: User Cancels Editing** 1. At step 3 or 4, the user clicks "Cancel". 2. System discards changes and closes modal without saving.
Exception flows	**EF-01: Invalid Task Data** 1. At step 5, the system detects an empty title or missing category. 2. The system displays an inline error message and prevents saving.
Business rules	BR-01, BR-03, BR-04, BR-05
Non-functional constraints	NFR-06 (UI updates within ≤ 200 ms)

7.3 Use Case UC-03

Use case ID	UC-03
Use case name	Delete task
Brief description	Allows the user to permanently remove a selected task after confirming the deletion.
Primary actor	User
Secondary actors	Supabase/ local storage
Priority	Must
Related requirements	FR-04, FR-06, NFR-06, NFR-07
Trigger	The user selects the "Delete" icon on a task item.
Pre-conditions	The selected task exists and is displayed on the task dashboard.
Post-conditions	1. The task is removed from local storage and queued for deletion on Supabase. 2. Daily progress percentage and progress bar are recalculated.
Main flow	1. The user clicks the "Delete" icon on a task item. 2. The system displays a confirmation dialog asking the user to confirm deletion. 3. The user clicks "Confirm". 4. The system removes task records from local storage. 5. The system queues cloud database removal if cloud sync is active. 6. The system refreshes dashboard UI and updates progress bar.
Alternative flows	**AF-01: User Cancels Deletion** 1. At step 3, the user clicks "Cancel" in the confirmation dialog.

	2. The system closes the dialog form without deleting the task.
Exception flows	None
Business rules	BR-07, BR-10
Non-functional constraints	NFR-06 (UI updates within ≤ 200 ms)

7.4 Use Case UC-04

Use case ID	**UC-04**
Use case name	Complete/ Revert task
Brief description	Toggles a task's status between Active and Completed, automatically updating daily progress metrics.
Primary actor	User
Secondary actors	Supabase/ Local storage
Priority	Must
Related requirements	FR-05, FR-06, FR-18, NFR-06
Trigger	The user clicks the completion checkbox next to a task.
Pre-conditions	The task is displayed on the active day's task list.
Post-conditions	1. Task status changes from Active to Completed (or vice versa). 2. Progress percentage and progress bar update immediately.
Main flow	1. The user clicks the status checkbox/toggle next to a task. 2. The system toggles status (Active to Completed, or Completed to Active). 3. The system saves updated status state to local storage automatically. 4. System recalculates daily progress percentage and updates progress bar UI.

Alternative flows	None
Exception flows	None
Business rules	BR-02, BR-07
Non-functional constraints	NFR-06 (UI reflects change in ≤ 200 ms)

7.5 Use Case UC-05

Use case ID	**UC-05**
Use case name	Set task priority
Brief description	Assigns a priority level (High, Medium, Low) to a task, displaying it with a designated color tag.
Primary actor	User
Secondary actors	Local storage
Priority	Must
Related requirements	FR-09, FR-10, NFR-06
Trigger	The user selects a priority option during task creation, edit, or direct list view interaction.
Pre-conditions	Task creation/edit window is active or task item is selected.
Post-conditions	The priority level is assigned to the task record and displayed with a proper visual color indicator.
Main flow	1. The user selects the priority level (High, Medium, or Low). 2. System assigns priority attribute to task record. 3. The system attaches a visual color tag: Red (High), Yellow (Medium), Green (Low). 4. System persists change to local storage and refreshes task UI.
Alternative flows	None

Exception flows	None
Business rules	BR-03
Non-functional constraints	NFR-06 (UI reflects change in ≤ 200 ms)

7.6 Use Case UC-06

Use case ID	**UC-06**
Use case name	View daily task list
Brief description	Displays all tasks associated with a selected date and allows switching between different dates.
Primary actor	User
Secondary actors	Local storage
Priority	Must
Related requirements	FR-07, FR-11, FR-12, NFR-01
Trigger	The user opens the application or selects a specific date on the date navigation bar.
Pre-conditions	The application is running and the user is in the main dashboard screen.
Post-conditions	Tasks assigned to the selected date are retrieved and displayed on screen.
Main flow	1. The user selects the desired date using date switcher controls. 2. System queries local storage for tasks matching selected date. 3. The system displays the task assigned to the selected date.
Alternative flows	**AF-01: No Tasks for Selected Date** 1. At step 2, the query returns zero matching tasks. 2. The system displays an empty state view (*"No tasks*

	scheduled for this day").
Exception flows	None
Business rules	BR-05, BR-06
Non-functional constraints	NFR-01 (Application loads task list within ≤ 3 s at startup)

7.7 Use Case UC-07

Use case ID	**UC-07**
Use case name	Filter task
Brief description	Filters visible task list by status (All, Active, Completed), priority (High, Medium, Low), and category simultaneously.
Primary actor	User
Secondary actors	None
Priority	Must
Related requirements	FR-13, FR-14, FR-15, FR-16, NFR-04, NFR-05
Trigger	The user selects options in status, priority, or category filter controls.
Pre-conditions	The task dashboard is open and displaying tasks.
Post-conditions	The dashboard displays only tasks matching all active filter criteria.
Main flow	1. The user opens the filter panel on the main dashboard by clicking on the filter icon/ button. 2. User selects filter parameters (e.g., Status: Active; Priority: High; Category: Work). 3. System evaluates task collection using selected combined criteria. 4. System updates task list display.
Alternative flows	**AF-01: Reset Filters** 1. The user selects "Clear/Reset Filters".

	2. The system removes active criteria and displays a full daily task list.
Exception flows	None
Business rules	BR-02, BR-03, BR-04
Non-functional constraints	NFR-05 (Filter updates list in ≤ 300 ms for up to 1,000 tasks), NFR-04

7.8 Use Case UC-08

Use case ID	**UC-08**
Use case name	View daily progress
Brief description	Calculates and displays daily task completion progress as a rounded percentage and visual progress bar.
Primary actor	User
Secondary actors	None
Priority	Must
Related requirements	FR-17, FR-18, NFR-06
Trigger	Application displays main dashboard or task state changes (added, completed, reverted, deleted).
Pre-conditions	Active date task list is loaded.
Post-conditions	Visual progress bar and percentage display accurate real-time completion status.
Main flow	1. The system counts total non-deleted tasks and completed tasks for selected days. 2. System calculates progress percentage using formula: $(\ CompletedTasks \div TotalTasks\ ) \times 100$ rounded to the nearest whole percent. 3. The system renders progress percentage text and fills progress bar accordingly.

	4. The system dynamically re-renders indicators whenever a task is added, deleted, completed, or reverted.
Alternative flows	**AF-01: Zero Tasks Scheduled** 1. If total tasks for day equals 0, the system displays progress as 0% with an empty progress bar.
Exception flows	None
Business rules	BR-07
Non-functional constraints	NFR-06 (UI updates within ≤ 200 ms)

7.9 Use Case UC-09

Use case ID	**UC-09**
Use case name	Synchronize Task data
Brief description	Synchronizes local task modifications with the cloud database (Supabase) when internet connectivity is established.
Primary actor	User/ Supabase
Secondary actors	Supabase/ local storage
Priority	Should have
Related requirements	FR-21, FR-22, FR-23, FR-24, NFR-08, NFR-09
Trigger	Internet connection is restored, application starts with active network, or user authenticates.
Pre-conditions	1. The user is signed into an active account. 2. Internet connection is established.
Post-conditions	1. Local offline queue changes are uploaded to Supabase. 2. Remote database updates are pulled to local storage.
Main flow	1. System detects active network connection to Supabase.

	2. The system checks the offline queue for pending local task changes. 3. The system transmits pending offline changes to the cloud database. 4. The system retrieves updated remote task records and updates local storage. 5. System refreshes UI to display synchronized state.
Alternative flows	None
Exception flows	**EF-01: Data Conflict Resolution** 1. At step 3 or 4, the system detects conflicting versions of the same task record. 2. The system compares last-updated timestamps of both records. 3. System applies Last-Write-Wins rule, preserving version with latest timestamp.
Business rules	BR-09, BR-10
Non-functional constraints	NFR-08 (Must-haves function offline with 0 failures), NFR-09 (Sync completes without data loss or duplicate tasks)

8 Prototype / Mockups

[Present the screen list and wireframes/mockups for the main use cases (Figma, Penpot, Balsamiq, ...). Include the link to the interactive prototype. Check basic accessibility: contrast, labels, keyboard use.]

Screen	Related use case(s)	Mockup / link
Daily Task Dashboard	UC-06 View Daily Task List, UC-07 Filter Tasks, UC-08 View Daily Progress	Figma Mobile link Figma PC link
Create / Edit Task	UC-01 Create Task, UC-02 Edit Task, UC-05 Set Task Priority	Figma Mobile link Figma PC link
Task Details / Management	UC-03 Delete Task, UC-04 Complete / Revert Task, UC-05 Set Task Priority	Figma Mobile link Figma PC link
Synchronization / Settings	UC-09 Synchronize Task Data	Figma Mobile link Figma PC link

9 Requirements Traceability Matrix

[This matrix is extended in PA3 (design elements) and PA4 (test cases).]

Req. ID	Use case	Screen	Design element (PA3)	Test cases (PA4)
RTM -01	**UC-01** (Create Task)	Main Dashboard / Create Task Modal	*(TBD in PA3)*	*(TBD in PA4)*
RTM	**UC-02** (Edit	Task Detail	*(TBD in PA3)*	*(TBD in PA4)*

-02	Task)	Screen / Edit Task Modal		
RTM -03	**UC-03** (Delete Task)	Main Dashboard / Task Item Action	*(TBD in PA3)*	*(TBD in PA4)*
RTM -04	**UC-04** (Complete / Revert Task)	Main Dashboard / Task Item Checkbox	*(TBD in PA3)*	*(TBD in PA4)*
RTM-05	**UC-05** (Set Task Priority)	Create/Edit Task Mode	*(TBD in PA3)*	*(TBD in PA4)*
RTM -06	**UC-06** (View Daily Task List)	Main Dashboard Screen	*(TBD in PA3)*	*(TBD in PA4)*
RTM -07	**UC-07** (Filter Tasks)	Main Dashboard Screen (Filter Bar)	*(TBD in PA3)*	*(TBD in PA4)*
RTM -08	**UC-08** (View Daily	Main Dashboard	*(TBD in PA3)*	*(TBD in PA4)*

	Progress)	Screen (Progress Bar)		
RTM -09	UC-09 (Synchronize Task Data)	Settings Screen / Background Sync Service	*(TBD in PA3)*	*(TBD in PA4)*

10 Requirements Review

[Record the requirements review held before submission (who took part, date, issues found and how they were resolved). This version becomes the requirements baseline.]

#	Issue found	Requirement(s)	Resolution
1	In section 5.1, functional requirements related category filtering was missing	5.1 Functional Requirements	After discussion with the BA, one functional requirement related to category filtering was added
2	In section 5.2, Non-Functional requirement regarding the usability of the filtering feature was missing	5.2 Non-Functional Requirements	After discussion with the BA, one Non-Functional requirement detailing the usability of the filtering feature was added
3	In section 5.3, domain requirement about how many categories can a task fall under was missing	5.3 Domain Requirements and Business Rules	After discussion with the BA, one Domain Requirement about the limit on the number of categories can a task fall under
4	In section 6.3, many of the functional requirements listed in the "Related FR" column did not match with the use cases	6.3 Use Case List	After another comparison with the listed functional requirements in section 2.1. The "Related FR" column was updated to better match the use case
5	In section 8, the list of screens for the PC platform was not listed in the table nor were the links to the interactive prototypes provided. Only the screenshots of the screens were present which are not enough for submission	8 Prototype/ Mockups	Notify the one responsible for this section about the mistake so that they can fix it

11 AI Usage Log

[Required in every PA (course GenAI policy). List each significant use of AI tools. Write "No AI tools were used" if none. Undisclosed AI use is treated as an academic integrity violation.]

#	Member	Tool	Task / purpose	How the output was verified and changed
1	Nguyễn Lương Hoàng Duy	Gemini	Reviewing and refining the drafts for sections 2.1 and 3. In section 5, identify requirements that do not fit the definitions of functional, non-functional and domain requirement. In section 7, based on the list of requirements and use cases, draft the use case specification for all use cases.	The drafts for sections 2.1 and 3 after AI review and refinement was finalized. No requirements that do not fit the definitions of functional, non-functional and domain requirements were identified. After reading through all use case specifications, no inconsistencies with the list of requirements and use case were found.
2	Nguyễn việt Quang	ChatGPT, FigmaAI	FigmaAI: To generate a mobile UI prototype for the Task Planner application, including the calendar, daily task list, task creation, and progress indicator. ChatGPT: To help organize the	FigmaAI: the generated prototype was compared with the traceability table to check whether the relevant screens and UI components supported the specified use cases. The layout and UI elements were reviewed and adjusted where necessary to better align with the

			traceability table linking functional requirements (FR-01 to FR-09), use cases (UC-01 to UC-09), and corresponding UI screens or components.	functional requirements. ChatGPT: The generated table was reviewed to ensure that each functional requirement was mapped to the corresponding use case and relevant UI screen or component. The mappings were checked against the project's requirements, and the table was adjusted where necessary to reflect the intended application features. .
3	Niê Ngọc Lâm	Claude	Drafting section 5, checking it for conflicts with other sections, and keeping the information up to date.	Content was checked manually; AI was used to check the logic and fix inconsistencies.
4	Trần Phúc Đạt	ChatGPT FigmaAI	ChatGPT: Assisted with drafting Section 6, Use Case Model, including the use case diagram, actors, and use case list based on the WhatYaDoing? application requirements. Figma AI: Used to create and refine the PC prototype/mockups for Section 8, including the daily task dashboard, task creation/editing, task management popups,	The use cases and actors were reviewed against the application's features and requirements. The PC mockups were reviewed against the existing mobile design and adjusted to keep the same overall design while fitting the desktop platform. The prototype was also checked to ensure the main use cases could be demonstrated through the interactive screens.

			filtering, synchronization, and settings.	